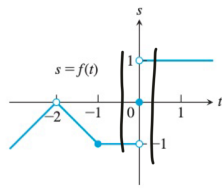


2. For the function $f(t)$ graphed here, find the following limits or explain why they do not exist.

- a. $\lim_{t \rightarrow -2} f(t)$ b. $\lim_{t \rightarrow -1} f(t)$ c. $\lim_{t \rightarrow 0} f(t)$ d. $\lim_{t \rightarrow 0.5} f(t)$



(2)

(a)

$$\lim_{t \rightarrow -2^+} f(t) = \lim_{t \rightarrow -2^-} f(t) = 0 \neq$$

(b)

$$\lim_{t \rightarrow -1^-} f(t) = \lim_{t \rightarrow -1^+} f(t) = -1 \neq$$

(c)

$$\therefore \lim_{t \rightarrow 0^-} f(t) = -1 \neq \lim_{t \rightarrow 0^+} f(t) = 1$$

$$\therefore \lim_{t \rightarrow 0} f(t) \text{ DNE}$$

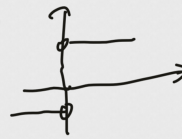
(d)

$$\lim_{t \rightarrow 0.5^-} f(t) = \lim_{t \rightarrow 0.5^+} f(t) = -1 \neq$$

7. Suppose that a function $f(x)$ is defined for all real values of x except $x = c$. Can anything be said about the existence of $\lim_{x \rightarrow c} f(x)$? Give reasons for your answer.

8. Suppose that a function $f(x)$ is defined for all x in $[-1, 1]$. Can anything be said about the existence of $\lim_{x \rightarrow 0} f(x)$? Give reasons for your answer.

(7) $f(x)$

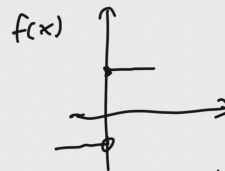


Does not guarantee that there's no jump about c

, take $c=0$ for $f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$ for counterexample

$$\therefore \lim_{x \rightarrow 0^-} f(x) = -1 \neq \lim_{x \rightarrow 0^+} f(x) = 1 \therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

(8)



Does not guarantee that there's no jump at 0, take

$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$ for example

$$\therefore \lim_{x \rightarrow 0^-} f(x) = -1 \neq \lim_{x \rightarrow 0^+} f(x) = 1 \therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

Finding Deltas Algebraically

Each of Exercises 15–30 gives a function $f(x)$ and numbers L , c , and $\varepsilon > 0$. In each case, find an open interval about c on which the inequality $|f(x) - L| < \varepsilon$ holds. Then give a value for $\delta > 0$ such that for all x satisfying $0 < |x - c| < \delta$ the inequality $|f(x) - L| < \varepsilon$ holds.

19. $f(x) = \sqrt{19 - x}$, $L = 3$, $c = 10$, $\varepsilon = 1$

29. $f(x) = mx + b$, $m > 0$, $L = (m/2) + b$,
 $c = 1/2$, $\varepsilon = c > 0$

(19)

$$-\varepsilon < f(x) - L < \varepsilon$$

$$L - \varepsilon < f(x) < L + \varepsilon$$

$$2 < f(x) < 4$$

$$4 < 19 - x < 16$$

$$-15 < -x < -3$$

$$\Rightarrow 3 < x < 15$$

$\Rightarrow$ The distance to c can be at most 5
for $|f(x) - L| < \varepsilon$ to hold

$$\Rightarrow \text{let } \delta = 5$$

(29)

$$L - \varepsilon < f(x) < L + \varepsilon$$

$$L - \varepsilon < mx + b < L + \varepsilon$$

$$\frac{L - \varepsilon - b}{m} < x < \frac{L + \varepsilon - b}{m}$$

$$\frac{L - \varepsilon}{m} < x < \frac{L + \varepsilon}{m}$$

$$\frac{1}{2} - \frac{\varepsilon}{2m} < x < \frac{1}{2} + \frac{\varepsilon}{2m}$$

$$\frac{1}{2} - \frac{1}{2m} < x < \frac{1}{2} + \frac{1}{2m}$$

$$\Rightarrow |x - \frac{1}{2}| < \frac{1}{2m}$$

$$\Rightarrow \delta = \frac{1}{2m}$$

31. $\lim_{x \rightarrow 1} \frac{x^{-1} - 1}{x - 1}$

32. $\lim_{x \rightarrow 0} \frac{\frac{1}{x-1} + \frac{1}{x+1}}{x}$

56. Suppose that $\lim_{x \rightarrow -2} p(x) = 4$, $\lim_{x \rightarrow -2} r(x) = 0$, and $\lim_{x \rightarrow -2} s(x) = -3$. Find

a. $\lim_{x \rightarrow -2} (p(x) + r(x) + s(x))$

b. $\lim_{x \rightarrow -2} p(x) \cdot r(x) \cdot s(x)$

c. $\lim_{x \rightarrow -2} (-4p(x) + 5r(x))/s(x)$

(56)

(a) $4 + 0 + (-3) = 1$

(b) $4 \cdot 0 \cdot (-3) = 0$

(c) $(-16 + 0)/(-3) = \frac{16}{3}$

(31)

$$\frac{\frac{1}{x} - 1}{x - 1}$$

$$= \frac{-(x-1)}{x(x-1)}$$

$$= \frac{-1}{x}$$

$$= -1$$

(32)

$$\frac{x+1 + x-1}{x(x-1)(x+1)}$$

$$= \frac{2}{(x-1)(x+1)}$$

$$= -2$$

65. a. It can be shown that the inequalities

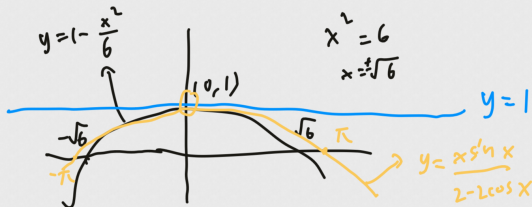
$$1 - \frac{x^2}{6} < \frac{x \sin x}{2 - 2 \cos x} < 1$$

hold for all values of x close to zero. What, if anything, does this tell you about

$$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}?$$

Give reasons for your answer.

T b. Graph $y = 1 - (x^2/6)$, $y = (x \sin x)/(2 - 2 \cos x)$, and $y = 1$ together for $-2 \leq x \leq 2$. Comment on the behavior of the graphs as $x \rightarrow 0$.



all three graphs approach 1 at $x \rightarrow 0$

(65) (a)

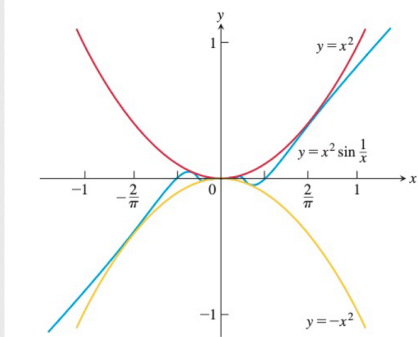
$$\therefore \lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = 1, \quad \lim_{x \rightarrow 0} 1 = 1$$

according to the squeeze theorem

$$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x} = 1$$

(b)

$$50. \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$



$$(50) \quad \therefore -1 \leq \sin \frac{1}{x} \leq 1$$

$$\therefore -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

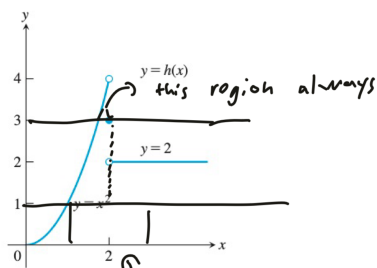
$$\lim_{x \rightarrow 0} -x^2 \leq \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} \leq \lim_{x \rightarrow 0} x^2$$

$$0 \leq \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} \leq 0$$

$\Rightarrow$ According to the squeeze theorem

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

$$58. \text{ Let } h(x) = \begin{cases} x^2, & x < 2 \\ 3, & x = 2 \\ 2, & x > 2 \end{cases}$$



Show that

a. $\lim_{x \rightarrow 2} h(x) \neq 4$

b. $\lim_{x \rightarrow 2} h(x) \neq 3$

c. $\lim_{x \rightarrow 2} h(x) \neq 2$

58

(a) $f(x) = h(x)$

$$0 < |x - 2| < \delta$$

$$2 - \delta < x < 2 + \delta$$

$$\text{suppose } \varepsilon = \frac{1}{2}, \quad x = 2 + \frac{\delta}{2}$$

$$\therefore |f(2 + \frac{\delta}{2}) - 4| = 2 \geq \frac{1}{2} \quad \text{AND} \quad 0 < (x - 2) < \delta$$

$$\therefore \lim_{x \rightarrow 2} f(x) \neq 4$$

(b) suppose $\varepsilon = \frac{1}{2}, \quad x = 2 + \frac{\delta}{2}$

$$\therefore |f(2 + \frac{\delta}{2}) - 3| = 1 \geq \frac{1}{2} \quad \text{AND} \quad 0 < (x - 2) < \delta$$

$$\therefore \lim_{x \rightarrow 2} f(x) \neq 3$$

(c) suppose $\varepsilon = 1, \quad \text{let } x = 2 - \min(2 - \sqrt{3}, \frac{\delta}{2})$

$$\Rightarrow f(x) \geq 3$$

$$\Rightarrow |f(x) - 2| \geq 1 = \varepsilon$$

$$\Rightarrow \lim_{x \rightarrow 2} f(x) \neq 2$$